

	Orden	No Orden
1	✓	
2	✓	
3	✓	
4	✓	
5		✓
6		✓
7	✓	
8	✓	
9	✓	✓
10		✓

① $L = \{ \text{Palíndromos de } a, b \}$

$$S \rightarrow aSa \mid bSb \mid a \mid b \mid \lambda$$

② $L = \{ a^i b^j c^k \mid i=j \text{ o } j=k, i, j, k > 0 \}$

$$S \rightarrow A \mid B$$

$$A \rightarrow E \cdot F$$

$$E = aEb \mid ab$$

$$F = cF \mid c$$

$$A \rightarrow i=j$$

$$B \rightarrow GH$$

$$G = aG \mid a$$

$$H = bHc \mid bc$$

③ $\{ a^i b^i a^j b^j \mid i, j > 0 \}$

$$S \rightarrow A \mid B$$

$$A \rightarrow aAb \mid ab$$

$$B \rightarrow aAb \mid ab$$

④ $L = \{ a^i b^j c^k \mid k = i + 2j \}$

$$S \rightarrow \lambda \mid A$$

$$A \rightarrow aAc \mid B \mid \lambda$$

$$B \rightarrow bBcc \mid bcc \mid \lambda$$

⑤ $L = \{ (a+b)^n \mid N(b) \geq N(a) \}$

$$S \rightarrow SaSaSbS \mid SaSbSaS \mid SbSaSaS \mid \lambda$$

⑥ $L = (0+1)^* \mid N(0) \geq N(1) + 1$

$$S \rightarrow A0A$$

$$A \rightarrow A0A1A \mid A1A0A$$

**Nota: Aquí no valen las combinaciones
contraejemplo:*

bbaaaaaabb

⑦ $L = \{ a^i (b+c)^k \mid k > i \}$

$$S \rightarrow aSX \mid X$$

$$X \rightarrow b \mid c$$